HOSPITAL OPD QUEUE & ER BED ALLOCATION ENGINE REFERENCES AND LIBRARIES

Simulation Platform:

- Wokwi Online Simulator -https://wokwi.com/projects/477869947689290753

Programming Language:

- Arduino C++ for microcontroller programming.

Libraries Used:

1. **Wire.h Library**:

The Wire.h library is a standard Arduino library used for I²C (Inter-Integrated Circuit) communication. I²C is a serial communication protocol that allows electronic devices to communicate with a microcontroller using two communication lines: SDA (Serial Data) and SCL (Serial Clock).

In the Hospital OPD Queue and ER Bed Allocation Engine project, the Wire.h library supports communication between the Arduino board and the I²C LCD module. The LCD is used to display patient registration details, token numbers, queue status, emergency information, and bed allocation results.

The I²C communication protocol reduces the number of wires required to connect peripheral devices. This makes the circuit simpler, more organized, and easier to implement. The Wire library provides functions to initialize I²C communication and transfer data between connected devices.

Important Functions:

- Wire.begin() – Initializes I²C communication.
- Wire.beginTransmission() – Starts communication with a device at a specified I²C address.
- Wire.write() – Sends data to the connected device.
- Wire.endTransmission() – Ends the data transmission.

Importance in the Project:

The Wire library supports the communication required by the I²C LCD interface. It helps the Arduino send information to the display, enabling users to view important details about patient registration and hospital management operations.

2. **LiquidCrystal_I2C.h Library:**

The LiquidCrystal_I2C.h library is used to control a Liquid Crystal Display (LCD) connected to an Arduino through an I²C interface. An LCD is an output device that displays text, numbers, and other information generated by a microcontroller.

In this project, a 20×4 I²C LCD is used to display patient details, token numbers, queue information, emergency priorities, and bed availability. A 20×4 LCD contains four rows, with each row capable of displaying up to 20 characters.

This library simplifies LCD programming by providing predefined functions to initialize the display, clear the screen, position the cursor, control the backlight, and display messages. It eliminates the need to control the LCD through complicated low-level instructions.

The I²C interface also reduces wiring because the LCD communicates with the Arduino using the SDA and SCL lines, along with power and ground connections. The LCD module may use an I²C address such as 0x27, depending on the hardware configuration.

Important Functions:

- lcd.init() – Initializes the LCD.
- lcd.backlight() – Turns on the LCD backlight.
- lcd.clear() – Clears the displayed content.
- lcd.setCursor(column, row) – Sets the cursor position.
- lcd.print() – Displays text or numerical values on the LCD.

Importance in the Project:

The LiquidCrystal_I2C library provides a simple method for displaying patient registration information and the results of queue management and bed allocation operations. It improves the usability of the prototype by allowing users to view system information directly on the LCD without requiring a separate computer display.

3. **Keypad.h Library**:

The Keypad.h library is used to interface a matrix keypad with an Arduino board. A matrix keypad is an input device consisting of multiple keys arranged in rows and columns. It allows users to enter numerical values, characters, and commands into an embedded system.

In the Hospital OPD Queue and ER Bed Allocation Engine project, the keypad acts as an input device for interacting with the system. It can be used to enter patient information, select menu options, and perform registration and bed-management operations according to the program's design.

A matrix keypad reduces the number of Arduino pins required to connect multiple keys. Instead of connecting each key to an individual pin, the keys are arranged in rows and columns. The Arduino identifies a pressed key by scanning the row and column connections.

The Keypad library simplifies this process by providing functions to detect key presses and return the corresponding character to the program. This reduces programming complexity and makes keypad interfacing easier.

Important Functions:

- makeKeymap() – Defines the arrangement of keys in the keypad layout.
- keypad.getKey() – Checks for a key press and returns the corresponding character when a key is detected.
- Keypad() – Creates a keypad object using the key layout, row pins, column pins, and keypad dimensions.

Importance in the Project:

The Keypad library enables interaction between the user and the hospital management system. It provides a convenient way to enter information and select operations. When combined with the Arduino program and LCD, it helps create an interactive prototype for patient registration, queue management, and bed allocation.

4. **Combined Role of the Libraries**:

The three libraries work together to support the input and output operations of the Hospital OPD Queue and ER Bed Allocation Engine.

The Keypad.h library accepts user input through the keypad. The Arduino C++ program processes the entered information and executes the required operations, such as patient registration, token management, priority handling, and bed allocation. The Wire.h library supports I²C communication, while the LiquidCrystal_I2C.h library provides functions to control the LCD and display the processed information.

These libraries simplify hardware interfacing and reduce the amount of low-level code required to operate the input and output devices. Their combined use makes the prototype easier to operate and demonstrates how embedded systems can be used to support hospital queue management and emergency bed allocation.